

Math 514, F13

Quiz 8

Name: Answer

SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Suppose that you put \$1,000 in a bank account at time $t = 0$. The varying interest rate is $r(t) = 0.02(2 + \sin t)$ (t measured by year) and compounded continuously. Compute the average of the spot interest rate up to time t , $\tilde{r}(t)$, and the amount in your account after 1 year.

$$\begin{aligned}
 \tilde{r}(t) &= \frac{1}{t} \int_0^t r(s) ds = \frac{1}{t} \int_0^t 0.02(2 + \sin s) ds \\
 &= \frac{0.02}{t} [2s - \cos s]_{s=0}^{s=t} \\
 &= \frac{0.02}{t} (2t - \cos t + 1) \\
 D(t) &= 1000 e^{\tilde{r}(t) \cdot t} \Rightarrow D(1) = 1000 \cdot e^{\tilde{r}(1) \cdot 1} \\
 &= 1000 \cdot e^{0.02(2 - \cos 1 + 1)} \\
 &\approx 1048.72
 \end{aligned}$$

Problem 2. (4 points) Suppose you pay 10 to buy an option to purchase a share of stock at a fixed price $K = 100$ after 2 years. Assume the nominal annual interest rate is 6% and compounded continuously. Find the present value of your return from this investment if the price of the stock after 2 year is 110.

$$\begin{aligned}
 &(110 - 100)e^{-rt} - 10 \\
 &= 10 \cdot e^{-0.06 \cdot 2} - 10 \\
 &\approx -1.1308
 \end{aligned}$$